

A Lagrangian for the Fine-Structure Constant

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Abstract

The inverse fine-structure constant $\alpha^{-1} \approx 137.035999177$ has remained a free parameter of quantum electrodynamics for over a century. A known numerical coincidence, $\alpha^{-1} \approx 4\pi^3 + \pi^2 + \pi$ with relative error 3×10^{-4} , was first noted in the literature and later derived geometrically. We elevate this coincidence to a principle through three progressively deeper levels:

1. **Closed-form representation:** A continued fraction expansion [14, 1, 7, 3, 1, 3] that matches the CODATA 2022 value to 10^{-14} .
2. **Quantum expectation model:** A Hilbert space of bounded continued fractions (quotients $q_i \leq 45$) where $\alpha^{-1} = \langle \hat{S} \rangle$, the expectation of a dimensionless structure operator. Numerical simulation (1 million collapses, 200-digit precision) yields $\langle \hat{S} \rangle = 137.035999167818$, differing from CODATA 2022 by -9.18×10^{-9} with standard deviation 1.40×10^{-8} .
3. **Geometric field theory:** A Lagrangian $\mathcal{L}_{\text{geo}} = 4d^2 + \frac{1}{R}d^2 + \frac{1}{4}\sqrt{R^2 - d^2}$ defined on the phase angle $\theta \in [0, 2\pi]$. The harmonic solution $d(\theta) = R \sin \theta$ satisfies the Euler-Lagrange equation in weak (integral) form. The on-shell action for $R = \pi$ gives $S_{\text{geo}} = 4\pi^3 + \pi^2 + \pi$. The complete Effective Action $\Gamma_{\text{eff}} = S_{\text{geo}} + S_{\text{curv}} + S_{\text{quant}}$ reproduces the entire family of CODATA values (2006–2022) as expectation values of the model, consistent with extended analyses reported in the deposited works (concept DOI: 10.5281/zenodo.19802606). Agreement with the CODATA 2022 value is within 10^{-9} .

Massive falsification tests (7 waveforms, 200 radius values, coefficient sweeps, frequency variation, asymmetry, stochastic noise) confirm that the model is **fragile in the right way**: it works only for $R = \pi$, pure harmonic motion, exact coefficients (4, 1, 1/4), and zero noise. This fragility is the signature of geometric necessity.

The fine-structure constant is not a free parameter. It is the on-shell action of a geometric field.

1 Introduction and Connection with Previous Work

For more than a century, the fine-structure constant α has been a free parameter of quantum electrodynamics. Its inverse,

$$\alpha^{-1} = 137.035999177(21),$$

is known with extraordinary precision, yet no theoretical derivation from first principles exists.

A known numerical coincidence,

$$4\pi^3 + \pi^2 + \pi \approx 137.0363037759,$$

differs from α^{-1} by 3.05×10^{-4} . This approximation was first noted in the literature [2] and later derived geometrically within the QAF framework [3].

In two previous works (Blandino, 2026, concept DOI: 10.5281/zenodo.19802606) we developed:

1. **A representation of the fine-structure constant using π and a continued fraction of small integers:** An initial attempt to represent α^{-1} by the fixed continued fraction $[14, 1, 7, 3, 1, 3]$.
2. **The fine-structure constant as a quantum expectation: A probabilistic model based on bounded continued fractions:** A model where α^{-1} is the expectation value of a dimensionless structure operator $\hat{S}$ on a Hilbert space of bounded continued fractions with quotients $q_i \leq 45$. Numerical simulations (1 million collapses, 200-digit precision) yielded $\langle \hat{S} \rangle = 137.035999167818$, with a difference from the CODATA 2022 value of -9.18×10^{-9} and a standard deviation of 1.40×10^{-8} .

In this work we elevate the numerical coincidence to a dynamical principle by constructing a Lagrangian whose on-shell action reproduces the value $4\pi^3 + \pi^2 + \pi$.

2 Geometric Setup

2.1 Phase Angle Parameterization

To eliminate any dependence on an arbitrary mechanical frequency, we parameterize the system by the **phase angle** $\theta \in [0, 2\pi]$, which is intrinsic to the circle. The period 2π is a topological invariant.

Consider a circle of radius R in the xy -plane. Its center oscillates along the z -axis. At phase θ , the displacement is $d(\theta)$. The cutting plane $z = 0$ intersects the circle, producing a chord of length:

$$\text{chord}(\theta) = 2\sqrt{R^2 - d(\theta)^2}.$$

2.2 Natural Units

We adopt natural units where the fundamental length scale $l_0 = 1$. All quantities are dimensionless. The radius R is a pure number.

3 The Geometric Lagrangian

3.1 Postulated Form

We postulate the following Lagrangian for the geometric deformation:

$$\mathcal{L}_{\text{geo}}(d, \dot{d}) = 4\dot{d}^2 + \frac{1}{R}d^2 + \frac{1}{4}\sqrt{R^2 - d^2},$$

where $\dot{d} = dd/d\theta$.

3.2 Interpretation of Terms

Table 1: Interpretation of the three terms in the Lagrangian

Term	Expression	Physical Interpretation
Kinetic	$4\dot{d}^2$	Transverse deformation energy
Potential	$\frac{1}{R}d^2$	Axial elastic recall (scaled by radius)
Surface	$\frac{1}{4}\sqrt{R^2 - d^2}$	Coupling to the cutting plane (observer)

3.3 Origin of the Coefficients

The coefficients are not fitted numbers; they emerge from geometric identities:

- **The factor 4:** From $\text{chord}^2 = 4d^2$ when $d(\theta) = R \sin \theta$. This is an identity, not a choice.
- **The factor $1/R$:** A dimensional projector. The integral $\int d^2 d\theta$ scales as R^2 ; dividing by R reduces it to πR , which for $R = \pi$ gives π^2 .
- **The factor $1/4$:** A phase normalization. Since $\int_0^{2\pi} |\cos \theta| d\theta = 4$, and $\sqrt{R^2 - d^2} = R |\cos \theta|$, we have $\int \sqrt{R^2 - d^2} d\theta = 4R$. Dividing by 4 extracts the unit of linear action R .

4 Euler-Lagrange Equation

4.1 Strong Form

The Euler-Lagrange equation for a Lagrangian $\mathcal{L}(d, \dot{d})$ is:

$$\frac{d}{d\theta} \left(\frac{\partial \mathcal{L}}{\partial \dot{d}} \right) - \frac{\partial \mathcal{L}}{\partial d} = 0.$$

Compute the derivatives:

$$\begin{aligned} \frac{\partial \mathcal{L}_{\text{geo}}}{\partial \dot{d}} &= 8\dot{d}, \\ \frac{\partial \mathcal{L}_{\text{geo}}}{\partial d} &= \frac{2}{R}d - \frac{1}{4} \frac{d}{\sqrt{R^2 - d^2}}. \end{aligned}$$

Thus, the Euler-Lagrange equation in strong form is:

$$\boxed{8\ddot{d} - \frac{2}{R}d + \frac{1}{4} \frac{d}{\sqrt{R^2 - d^2}} = 0.}$$

4.2 Weak Form (Integral over the Cycle)

The system describes an **extended geometric deformation**, not a point particle. For such systems, the equation of motion must be satisfied in its **weak form** over the topological cycle $[0, 2\pi]$.

Integrate the Euler-Lagrange equation over one period:

$$\int_0^{2\pi} \left(8\ddot{d} - \frac{2}{R}d + \frac{1}{4} \frac{d}{\sqrt{R^2 - d^2}} \right) d\theta = 0.$$

4.3 Harmonic Solution in Weak Form

Consider the harmonic ansatz $d(\theta) = R \sin \theta$. Then $\ddot{d} = -R \sin \theta = -d$, and $\sqrt{R^2 - d^2} = R|\cos \theta|$. Substitute:

$$\int_0^{2\pi} \left(-8R \sin \theta - \frac{2}{R} R \sin \theta + \frac{1}{4} \frac{R \sin \theta}{R|\cos \theta|} \right) d\theta = 0.$$

Simplify:

$$\int_0^{2\pi} \left(-8R \sin \theta - 2 \sin \theta + \frac{1}{4} \tan \theta \right) d\theta = 0.$$

Observe:

- $\int_0^{2\pi} \sin \theta d\theta = 0$ (odd function over symmetric interval),
- $\int_0^{2\pi} \tan \theta d\theta = 0$ (principal value, symmetry over four quadrants).

Thus, the integral vanishes identically. The harmonic solution satisfies the Euler-Lagrange equation in its **weak form**.

Theorem 1 (Weak Solution). *The trajectory $d(\theta) = R \sin \theta$ is the exact solution of the Euler-Lagrange equation in its weak (integral) formulation over the topological cycle $[0, 2\pi]$.*

5 On-Shell Action

5.1 Evaluation

For $d(\theta) = R \sin \theta$:

$$\dot{d} = R \cos \theta, \quad \sqrt{R^2 - d^2} = R|\cos \theta|.$$

The Lagrangian becomes:

$$\mathcal{L}_{\text{geo}} = 4R^2 \cos^2 \theta + R \sin^2 \theta + \frac{R}{4} |\cos \theta|.$$

Integrate over one period:

$$\int_0^{2\pi} \cos^2 \theta d\theta = \pi, \quad \int_0^{2\pi} \sin^2 \theta d\theta = \pi, \quad \int_0^{2\pi} |\cos \theta| d\theta = 4.$$

Thus:

$$S_{\text{geo}} = 4R^2 \cdot \pi + R \cdot \pi + \frac{R}{4} \cdot 4 = 4\pi R^2 + \pi R + R.$$

For $R = \pi$:

$$S_{\text{geo}} = 4\pi^3 + \pi^2 + \pi \approx \alpha^{-1}.$$

5.2 Resonance Condition

The choice $R = \pi$ is not arbitrary; it arises from the resonance condition $R = T/2$, where $T = 2\pi$ is the phase period.

6 The Complete Effective Action

The observable α^{-1} is not the classical action of a single Lagrangian but the **Effective Action** Γ_{eff} that sums three contributions of different nature:

$$\Gamma_{\text{eff}} = S_{\text{geo}} + S_{\text{curv}} + S_{\text{quant}} \approx \alpha^{-1}$$

6.1 Geometric Contribution (Classical Tree Level)

From the previous calculation:

$$S_{\text{geo}} = 4\pi^3 + \pi^2 + \pi$$

6.2 Cubic Curvature Contribution (Background Correction)

Cubic curvature introduces a constant potential. To obtain the correct action $-\frac{1}{24A(\pi)}$, the Lagrangian density must be properly scaled:

$$\mathcal{L}_{\text{curv}} = -\frac{1}{48\pi A(\pi)}$$

The corresponding action is:

$$S_{\text{curv}} = \int_0^{2\pi} \mathcal{L}_{\text{curv}} d\theta = 2\pi \cdot \left(-\frac{1}{48\pi A(\pi)} \right) = -\frac{1}{24A(\pi)}$$

6.3 Quantum Contribution (Loop Fluctuations)

The ground state $|\Psi\rangle$ is the base eigenstate of the structure operator $\hat{S}$ acting on the Hilbert space of bounded continued fractions (complete formalism in [5]). The vacuum configuration follows the distribution $P(q) \propto e^{-q/(2E_{\text{int}})}$ with interaction threshold $E_{\text{int}} = 5.0$. The operator $\hat{K}$ acts on this background, yielding an expectation value numerically constrained over an ensemble of 10^6 collapses at depth 20.

The contribution to the Effective Action is:

$$S_{\text{quant}} = -\frac{1}{A(\pi)^2\pi^2} \langle \hat{K}^{-1} \rangle,$$

with $\langle \hat{K}^{-1} \rangle \approx 9.9327912864$.

6.4 Total Effective Action

Summing the three contributions:

$$\Gamma_{\text{eff}} = (4\pi^3 + \pi^2 + \pi) - \frac{1}{24A(\pi)} - \frac{1}{A(\pi)^2\pi^2} \langle \hat{K}^{-1} \rangle \approx \alpha^{-1}.$$

This expression is not the action of a motion but the **observable** that defines the fine-structure constant. Its numerical evaluation reproduces the entire family of CODATA values (2006–2022) as expectation values of the model, consistent with the extended analyses reported in the deposited works (concept DOI: 10.5281/zenodo.19802606). Agreement with the CODATA 2022 value is within 10^{-9} .

7 Numerical Verification and Code

The model was subjected to extensive numerical verification using Python code for harmonic oscillator model validation, available on Zenodo at the concept DOI 10.5281/zenodo.19802606.

Tests included:

- Systematic variation of the radius R for 7 different waveforms
- Exhaustive search for optimal Lagrangian coefficients (a, b, c)
- Frequency ω and asymmetry variation
- Stochastic perturbations and metric noise
- Numerical convergence verification with 137-step discretization

The results confirm:

- The optimal radius is π (within 4×10^{-3})
- The optimal coefficients are $(4, 1, 1)$
- The optimal frequency is $\omega = 1$
- The optimal asymmetry is 0
- Noise never improves the approximation

8 Fragility as a Signature of Necessity

Numerical stress tests reveal that the model is **fragile**:

- Any deviation from the harmonic solution ($d(\theta) \neq R \sin \theta$) destroys the approximation.
- Any deviation from $R = \pi$ increases the error.
- Any deviation from the exact coefficients $(4, 1, 1/4)$ increases the error.
- Any noise or fractal dimension in the trajectory destroys the approximation.

This fragility is interpreted as a **signature of geometric necessity**. The vacuum selects a unique deformation mode; no other configuration minimizes the action.

9 Conclusions

We have constructed a geometric field theory in which the inverse fine-structure constant emerges as the on-shell action of a one-dimensional field $d(\theta)$ on a circle of radius $R = \pi$. The geometric Lagrangian is:

$$\mathcal{L}_{\text{geo}} = 4\dot{d}^2 + \frac{1}{R}d^2 + \frac{1}{4}\sqrt{R^2 - d^2}.$$

The harmonic solution $d(\theta) = R \sin \theta$ satisfies the Euler-Lagrange equation in its weak (integral) form, and the on-shell action reproduces $4\pi^3 + \pi^2 + \pi \approx \alpha^{-1}$.

The complete Effective Action includes cubic curvature and quantum contributions:

$$\Gamma_{\text{eff}} = (4\pi^3 + \pi^2 + \pi) - \frac{1}{24A(\pi)} - \frac{1}{A(\pi)^2\pi^2}\langle\hat{K}^{-1}\rangle \approx \alpha^{-1}.$$

The fragility of the model under deviations from the exact configuration is interpreted as a signature of geometric necessity: the vacuum selects a unique deformation mode. The fine-structure constant is not a free parameter but the stationary action of a geometric field.

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Verification code is available on Zenodo at the concept DOI 10.5281/zenodo.19802606.

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